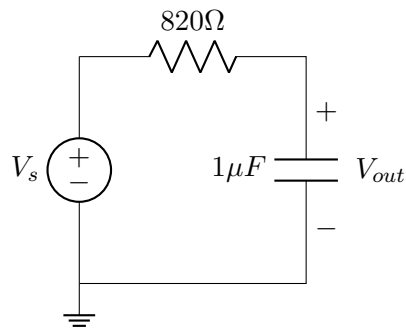


RC Circuit Response to Step, Pulse, and Square Inputs

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January 30, 2026

0.1 RC circuit diagram

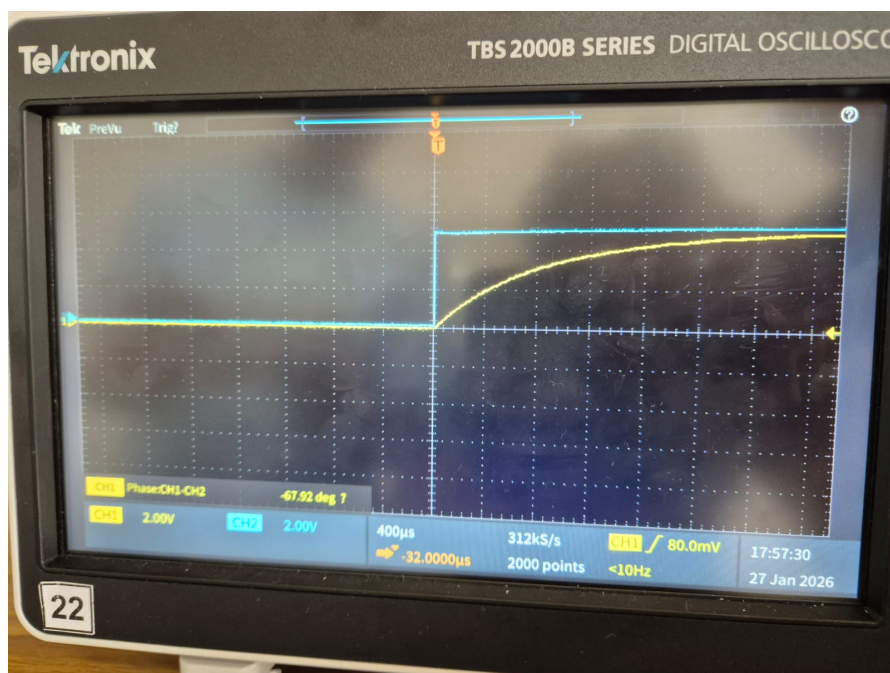


Here is the diagram of a RC circuit. RC circuits are low-pass filter circuits. The value of resistance and capacitance was chosen to be $R = 820\Omega$ and $C = 1\mu F$. We will now observe the behaviour and relation of V_{out} for different V_s . We will observe the response of different step, pulse and square inputs.

0.2 Step Input (Single Event)

0.2.1 Charging Response

Experimental Results



Signal Information

Blue Waveform (CH2): Step input (square rise edge). A step input of the form $V_s = 5u(t)$ was applied. This input signal could be generated by the AFG by applying a pulse signal of very very large width.

Yellow Waveform (CH1): Capacitor charging curve. This is the output and the voltage across the capacitor. We can see that this is the type of response we get for a step signal.

CH1 scale: 2.00 V/div

CH2 scale: 2.00 V/div

Time scale: 400 μs /division

0.2.2 Theoretical response

For a step input voltage $V_s(T)$: $V_C(t) = V_s (1 - e^{-t/RC})$

Computing time constant from measurement

For computing the time constant, we look at the charging waveform.

We compute time constant by measuring the time voltage across resistor takes to reach the voltage at one time constant.

Here $V_C(t)$ is the voltage across the capacitor at time t .

At one time constant $t = \tau = RC$:

$$V_C(\tau) = V_s(1 - e^{-1}) = 0.632V_s \quad (0.1)$$

$$V_C(\tau) = 0.632 \times 5 = 3.16V \quad (0.2)$$

From the graph we can take around 8 small divisions (1.6 divisions) to find the corresponding time. We find that the corresponding time is close to $800\mu s$ due to it being around 2 time divisions away from the axis.

$$t \approx 800\mu s \quad (0.3)$$

This aligns with the actual value of resistances and capacitances we took. The theoretical value of t is $t = 820\mu s$. The error is less than 3%.

Verification - Hand Calculation

When the input is removed: $V_C(t) = V_0 e^{-t/RC}$

Substituting $RC = 0.82 ms$:

$$V_C(t) = V_0 e^{-t/0.82ms} \quad (0.4)$$

At $t = \tau$:

$$V_C = 0.368V_0 \quad (0.5)$$

At $t = 5\tau = 4.1 ms$, the capacitor is almost fully discharged.

Verification Using 2τ

The theoretical time constant of the RC circuit is:

$$\tau = RC = 820 \times 1 \times 10^{-6} = 0.82 \text{ ms}$$

Hence,

$$2\tau = 2 \times 0.82 = 1.64 \text{ ms}$$

For a step input, the capacitor charging equation is:

$$V_C(t) = V_s \left(1 - e^{-t/\tau}\right)$$

At $t = 2\tau$:

$$V_C(2\tau) = V_s (1 - e^{-2})$$

$$V_C(2\tau) = V_s (1 - 0.135)$$

$$\boxed{V_C(2\tau) \approx 0.865 V_s}$$

From the experimental waveform, the capacitor voltage at approximately $t = 1.64 \text{ ms}$ was observed to be close to 86.5% of the final value.

Thus, the experimental result agrees well with the theoretical prediction, verifying the step response of the RC circuit.

0.3 Pulse Input (Single Event)

0.3.1 Short pulse ($T \ll RC$)

Signal Information

Hand Calculation

$$T_p \ll 820 \mu s \quad (0.6)$$

Peak capacitor voltage:

$$V_{max} = V_s \left(1 - e^{-T/RC}\right) \quad (0.7)$$

For $T \ll RC$:

$$V_{max} \approx \frac{T}{RC} V_s \quad (0.8)$$

0.3.2 Pulse width ($T \approx RC$)

Signal Information

Hand Calculation

From (0.7), it follows that

$$T \approx 820\mu s \quad (0.9)$$

$$V_{max} = V_s(1 - e^{-1}) = 0.632V_s \quad (0.10)$$

0.3.3 Long Pulse $T \gg RC$

Signal Information

Hand Calculation

From (0.7), it follows that

$$T \gg 820\mu s \quad (0.11)$$

$$V_{max} \approx V_s \quad (0.12)$$

We notice that, the capacitor charges almost completely during ON time and discharges exponentially during OFF time.

0.4 Square-Wave Input (Transient and Steady-State)

0.4.1 $RC \gg T$

$$T \ll RC \quad (0.13)$$

$$T \ll 0.82 ms \quad (0.14)$$

Hand Calculations

For capacitor charging:

$$V_C(t) = V_i + (V_f - V_i) \left(1 - e^{-t/RC}\right) \quad (0.15)$$

Since $t \leq \frac{T}{2} \ll RC$,

$$\frac{t}{RC} \ll 1$$

Using approximation:

$$e^{-t/RC} \approx 1 - \frac{t}{RC}$$

$$V_C(t) \approx V_i + (V_f - V_i) \frac{t}{RC}$$

Inference

The capacitor voltage varies almost linearly with time. Hence, the output waveform is approximately triangular.

RC circuit behaves as an integrator

0.4.2 Case 2: $RC \approx T$

$$T \approx RC = 0.82 \text{ ms}$$

Hand Calculations

During the charging interval:

$$V_C(t) = V_s \left(1 - e^{-t/RC}\right) \quad (0.16)$$

At the end of half cycle:

$$t = \frac{T}{2} \approx \frac{RC}{2}$$

$$V_C = V_s (1 - e^{-0.5})$$

$$V_C \approx 0.393V_s$$

During discharge:

$$V_C(t) = V_0 e^{-t/RC} \quad (0.17)$$

At $t = \frac{T}{2}$:

$$V_C = V_0 e^{-0.5} \approx 0.607V_0$$

Inference

The capacitor partially charges and partially discharges in each cycle. Both transient and steady-state components are present.

Maximum transient behavior is observed

THIS IS CASE THREE

0.4.3 Case 3: $RC \ll T$

$$T \gg RC$$

$$T \gg 0.82 \text{ ms}$$

Hand Calculations

During ON time:

$$t = \frac{T}{2} \gg RC$$

$$e^{-t/RC} \approx 0$$

$$V_C \approx V_s$$

During OFF time:

$$V_C(t) = V_s e^{-t/RC} \tag{0.18}$$

For $t \gg RC$:

$$V_C \approx 0$$

Inference

The capacitor fully charges and discharges within each cycle.

Output closely follows the input waveform